

What is an ingress controller?

- Web applications can have multiple microservices
- Exposing **each microservice individually** as a LoadBalancer Kubernetes **service** can be **cumbersome**
- **Kubernetes Ingress** allows you to **manage traffic** to your services by defining a set of rules that map incoming requests to the appropriate service
- **Ingress** provides a flexible and powerful way to **expose** your Kubernetes services **to the Internet** allowing managing the application **from a single point**
- An **Ingress Controller** is needed to **handle and enforce the Ingress rules** defined by the Ingress resource
- **Ingress Controller** is a software component that runs in the cluster and **listens for changes** to Ingress(es)
- **Ingress Controller updates its configuration** and **applies the new routing rules** to the underlying load balancer, application gateway or reverse proxy
- There are multiple **open-source ingress controllers** like **nginx**, traefix or haproxy
- **Azure** provides its **own ingress controller**, which is **Application Gateway Ingress Controller (AGIC)**
- Ingress and Ingress Controller provide a **layer 7 load balancer functionality**

